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High prevalence of Pantone-Valentine Leucocidin (PVL) toxin carrying MRSA and multidrug resistant gram negative bacteria in late onset neonatal sepsis indicate nosocomial spread in a Pakistani tertiary care hospital

Zainab Zahoor^{a,b}, Amna Mumtaz^a, Zia-ur-Rehman Farooqi^c, Noor Rehman^d, Nighat Batool^a, Zobia Noreen^a, Sadia Sattar^e, Nazish Bostan^e, Sundus Javed^{a,*}

^a Microbiology and Public Health Laboratory, Department of Biosciences, COMSATS University Islamabad, Park road, Tarlai Kalan, Islamabad, Pakistan

^b Faculty of Rehabilitation and Allied Health Sciences, Riphah International University, Islamabad, Pakistan

^c Khyber Medical University, Institute of Health Sciences, Peshawar, Pakistan

^d Department of Pathology, Khyber Teaching Hospital, Peshawar, Pakistan

^e Molecular Virology Lab, Department of Biosciences, COMSATS University Islamabad, Park road, Tarlai Kalan, Islamabad, Pakistan

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ABSTRACT

Background: Neonatal sepsis has high incidence with significant mortality and morbidity rates in Pakistan. We investigated common etiological patterns of neonatal sepsis at a tertiary care setup.

Methods: 90 pus and blood, gram negative and gram positive bacterial isolates were analyzed for virulence and antibiotic resistance gene profiling using PCR and disc diffusion methods.

Results: *Staphylococcus aureus* showed strong association with neonatal sepsis (43 %) followed by *Citrobacter freundii* (21 %), *Pseudomonas aeruginosa* (13 %), *Escherichia coli* (15 %) and *Salmonella enterica* (8 %). Molecular typing of *E. coli* isolates depicted high prevalence of the virulent F and B2 phylogroups, with 4 hypervirulent phylogroup G isolates. 76.9 % *S. aureus* isolates showed presence of *Luk-PV*, encoding for Pantone-valentine leucocidin (PVL) toxin with majority also carrying *MecA* gene and classified as methicillin resistant *S. aureus* (MRSA). *ecpA*, *papC*, *fimH* and *traT* virulence genes were detected in *E. coli* and *Salmonella* isolates. 47 % *Citrobacter freundii* isolates carried the shiga like toxin *StxII*. Antimicrobial resistance profiling depicted common resistance to cephalosporins, beta lactams and fluoroquinolones.

Conclusion: Presence of PVL carrying MRSA and multidrug resistant gram negative bacteria, all isolated from late onset sepsis neonates indicate a predominant nosocomial transmission pattern which may complicate management of the disease in NICU setups

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Introduction

Neonatal sepsis (NS) is a worldwide phenomenon that affects the life expectancy of neonates globally. According to an estimate 2202 newborns per 100,000 live births suffer from sepsis annually [1]. Neonatal sepsis leads to organ failure, tissue damage and ultimately results in neonatal death [2]. Etiology of neonatal sepsis can be varied due to variations in lifestyle and antibiotic use in different regions, therefore epidemiological pattern of neonatal sepsis shows

diversity [3]. Reports regarding disease burden of neonatal sepsis in terms of its morbidity in Pakistan show that 1–4 newborns among 1000 live births suffer from neonatal sepsis each year [4]. Khyber Pakhtunkhwa alone accounts for 62.5 % cases of neonatal sepsis while 42 % cases are from province of Sindh [4], indicating that these regions are the hotspots for NS in Pakistan. This may be due to poor neonatal care facilities in these regions.

Neonatal sepsis is broadly classified into two types, early onset neonatal sepsis (EOS, reported within three-seven days after birth) and late onset neonatal sepsis (LOS, reported after seven days of birth). It is proposed that the main cause of early onset neonatal sepsis is vertical transmission while late onset neonatal sepsis is caused due to nosocomial infections [5].

* Corresponding author.

E-mail address: sundus.javed@comsats.edu.pk (S. Javed).

The aim of this study was to isolate and identify bacterial pathogens associated with neonatal sepsis including *S. aureus*, *E. coli*, *C. freundii*, *P. aeruginosa* and *S. enterica*. We also aim to determine the antimicrobial resistance potential of isolates from septic patients and molecular analysis of virulence genes.

Methodology

Sample collection

Samples were collected from the hospital's Neonatal Intensive Care Unit (NICU) after obtaining necessary ethical approval for the study during month of April (2–April till 22–April 2021). A total of 150 samples were collected, both blood and pus, after taking consent from the guardian of the patient. Blood samples were collected in pediatric blood vial of Becton Dickinson and Company (BD BACTEC, USA) while pus samples were collected using sterile cotton swabs. Sample of only those patients was considered who were suffering from neonatal sepsis while patients hospitalized due to birth abnormalities or other congenital diseases were excluded. Blood samples were collected from suspected patient of neonatal sepsis through venipuncture into BD BACTEC blood culture bottle (pediatric). Culture bottles were incubated for 7 days in BACTEC 9240, Becton Dickinson and Company (BD BACTEC, USA). During these 7 days if bottles were detected positive for growth, they were taken from BACTEC 9240 and inoculated on Columbia blood agar (Oxoid, UK) and MacConkey media (Oxoid, UK). Pus sample was also taken from suspected patients using sterile cotton swabs. Cotton swabs were rubbed at site of cutaneous pus including limbs, forehead, and umbilical stalk. Swabs were inoculated on Columbia blood agar (Oxoid, UK) and MacConkey agar (Oxoid, UK). After an overnight incubation at 37° C bacterial growth was observed on culture plates. Pure bacterial growth was preserved in 15 % nutrient-glycerol stock solution and stored at – 20 °C.

Bacterial isolation and preliminary characterization

Culture media were prepared by following manufacturer's protocol. Preliminary characterization of bacterial isolates from septic neonates was made by observing growth of isolate on culture media, Gram staining and performing standardized biochemical tests. *S. aureus* grew on Mannitol Salt agar (MSA) (Oxoid, UK) as large yellow colonies. *S. enterica* grew as lactose non-fermenting pale colonies, on MacConkey agar (Oxoid, UK) Meanwhile, *C. freundii* grew as dry pink colonies on MacConkey agar as it ferments lactose. *Pseudomonas aeruginosa* grew as flat smooth colonies which have alligator's skin like appearance due to pigment production on MacConkey agar. While *E. coli* appear as pink or bright pink colonies on MacConkey agar. Biochemical tests including coagulase, catalase, DNase, oxidase, indole, urease, Sulfur, indole motility media (SIM), citrate and Triple sugar iron agar (TSI) tests were performed to determine bacterial identity as described earlier.

Antibiotic susceptibility profiling

Antibiotic resistance and susceptibility profile of the isolates was determined using the Kirby Bauer disc diffusion method. Commercially available antibiotic discs of azithromycin (ATM 15 µg), ceftazidime (30 µg), ceftriaxone (CRO 30 µg), nitrofurantoin (NIF 50 µg), cefaclor (CFC 30 µg), amoxicillin-clavulanic acid (AMC 20 +10 µg), gentamicin (CN 10 or 120 µg), cefuroxime (CXM 30 µg), ciprofloxacin (CIP 5 µg), penicillin (AST 6 µg), Piperacillin- tazobactam (TZP 75 +10 µg), Imipenem (IPM 10 µg) / meropenem (MEM 10 µg), levofloxacin (LVX 5 µg), amoxicillin (AMX 25 µg), tetracycline (TE 30 µg), tigecycline (TGC 15 µg), erythromycin (E 15 µg), vancomycin (V 5 µg), fosfomycin (FOS 50 µg), ampicillin (AMP 10 µg),

doxycycline (DO 30 µg), amikacin (AK 30 µg) and clindamycin (DA 2 µg) purchased from Oxoid, UK were used. The discs were applied to Mueller Hinton (MH) agar, (Oxoid, UK) plates inoculate with pure isolates in form of lawn culture. Results of antibiotic resistance and sensitivity were interpreted according to the criteria devised by Clinical and Laboratory Standard Institute (CLSI) Performance standards for Antimicrobial Susceptibility testing [6].

Molecular characterization of bacterial isolates

DNA extraction was performed using Solarbio® universal genomic DNA extraction kit (Solarbio, China) as per manufacturer instructions. Kit was used to extract DNA from *Salmonella* and *S. aureus*. For *E. coli* DNA was extracted by boiling method [7]. Conventional PCR was performed to identify bacterial species at molecular level. For this purpose, conserved 16s r-RNA gene of bacteria was targeted. PCR conditions and oligonucleotide primers used in this study are elaborated in in [supplementary table 1](#) respectively.

Virulence typing and phylogrouping of isolates

Virulence genes *LUK-PV* and *clbA*, *clbQ*, *ecpA*, *papC*, *fimH*, *traT*, *aggR* and *viaB*, *SlrIIB* were screened for *S. aureus* and *Enterobacteriaceae* respectively (list of primers along with their sequences is given in [supplementary table 2 and 3](#)).

Luk-PV gene of *S. aureus* was targeted as it encodes for a highly virulent toxin produced by *S. aureus* i.e., Pantone-valentine leucocidin toxin. Pantone-Valentine leucocidin (Luk-PV) attacks on white blood cells (leukocytes and neutrophils) and is characterized as pore forming toxin [8]. Multiplex PCR was performed in which screening of both genes *Luk-PV* and *mec-A* was done to quantify strains of *S. aureus* which were both virulent as well as methicillin resistant.

Phylogrouping is a quick and economical method as alternative to Multilocus sequence typing (MLST) to distinguish *E. coli* strains based on their virulence, niche and antibiotic resistance. To place all the isolated *E. coli* into specific phylogroup specific genes of *E. coli* *chuA* (288 bp), *yjaA* (211 bp), *TspE4. C2* (152 bp), *arpA* (400 bp) were amplified, using a multiplex PCR protocol described by Clermont and colleagues [9].

Strains of *E. coli* categorized into F and B2 phylogroups were further classified into phylogroup G by performing PCR, using *YbgD.G* primer [10].

Selected antibiotic resistance genes were screened among isolates to detect Extended spectrum beta lactamase (ESBL) and carbapenemase resistance in *E. coli* isolates. Multiplex PCR using *NDM*, *VIM*, *SIM*, *IMP* and *GIM* primers was performed as described earlier ([Supplementary table 4](#)) [11].

For *S. aureus*, presence of *mec-A* was screened. 1–1.5 % agarose gel was used to electrophore and visualize PCR product using UV transilluminator.

Statistical analysis

Microsoft Excel version 2201 was used for data compilation and statistical calculations. For qualitative data, we calculated percentage and median while for quantitative variables mean and standard deviation were estimated. Bar charts were made using GraphPad Prism version 9.3.1.

Results

Clinical characteristics of patients suffering from neonatal sepsis

Blood and pus samples were collected from 150 suspected patients of neonatal sepsis, Peshawar, while sample of patients suffering from neonatal sepsis in-addition to congenital diseases were

Table 1
Prevalence and clinical characteristics of patients suffering from neonatal sepsis.

Category	<i>Staphylococcus aureus</i> (n = 39)	<i>Citrobacter freundii</i> (n = 19)	<i>Pseudomonas aeruginosa</i> (n = 12)	<i>Escherichia coli</i> (n = 13)	<i>Salmonella enterica</i> (n = 7)
Early onset sepsis 3–7 days	14 (35.9 %)	3 (15.8 %)	3 (25 %)	3 (23.1 %)	1 (14.3 %)
Late onset sepsis > 7 days	25 (64.1 %)	16 (84.2 %)	9 (75 %)	10 (76.9 %)	6 (85.7 %)
Gender					
Male patients	29 (74.4 %)	14 (73.7 %)	6 (50 %)	8 (61.5 %)	4 (57.1 %)
Female patients	10 (25.6 %)	5 (26.3 %)	6 (50 %)	5 (38.5 %)	3 (42.9 %)
Type of sample/ Source					
Blood samples	18 (46.2 %)	5 (26.3 %)	0	5 (38.5 %)	7 (100 %)
Pus samples	21 (53.8 %)	14 (73.7 %)	12 (100 %)	8 (61.5 %)	0
Age					
Mean $\pm$ SD	20.18 $\pm$ 7.84	19.32 $\pm$ 9.22	24.85 $\pm$ 3.48	19 $\pm$ 7.08	22 $\pm$ 2.65
Median	21 days	20 days	25 days	20 days	22 days
Hospital stay					
0–7 days	14 (35.9 %)	3 (15.8 %)	2 (16.7 %)	3 (23.1 %)	1 (14.3 %)
> 7 days	25 (64.1 %)	16 (84.2 %)	10 (83.3 %)	10 (76.9 %)	6 (85.7 %)

excluded from study. Ninety out of the 150 patient samples showed growth of different bacterial pathogens and were categorized as culture-positive. These culture positive samples comprised of 55 taken from pus and 35 blood samples.

Bacterial pathogens were predominantly isolated from male neonates (61 neonates). Among these isolates 43 % were identified as *S. aureus* and 8 % *S. enterica*. Majority of isolates were obtained from patients aged 21–30 (51 %) days with isolation frequency highest from patients who had hospital stay of more than seven days (74.4 %) indicating a higher possibility of nosocomial spread (Table 1). *S. aureus* was isolated from sample of patients with mean age 20.18 $\pm$ 7.84 days. Similarly, *C. freundii* was isolated from sample of patients with mean age 19.32 $\pm$ 9.22 days, *P. aeruginosa* was bacterial pathogen commonly isolated from sample of patients with mean age 24.85 $\pm$ 3.48 days, *E. coli* was isolated from sample of patients with mean age 19 $\pm$ 7.08 days while *S. enterica* was frequently isolated from male patients with mean age 22 $\pm$ 2.65 (Table 1).

Molecular characterization of bacterial pathogens

The bacterial isolates that were phenotypically confirmed as *S. aureus*, *S. enterica*, *C. freundii* or *E. coli* were further subjected to PCR based identification using 16s rRNA gene targets specific for each bacterial species (primer sequences are elaborated in Supplementary Table 1). All seven isolates of *S. enterica* were therefore identified as *Salmonella typhi*. Similarly, all isolates were confirmed by PCR as either *S. aureus*, *C. freundii* or *E. coli* (Table 1).

E. coli was identified among 13 patients with 3 early and 10 cases of late hospital stay patients with a mean age of 19 $\pm$ 7.08 days. Phylogrouping was performed using the Clermont multiplex PCR method [6] to determine route of transmission as well as pathogenic mechanism of bacterial pathogen. Amongst the *E. coli* clinical isolates F, B2 and Clade III, IV or V phylogroups were observed most

frequently (Table 2), each having similar isolation frequency of 23.07 % (3/13) while only 1 isolate (7.6 %) belonged to phylogroup D or E. None of the isolates belonged to phylogroup A, B1 or C. Recent screening method described by Clermont and colleagues [6] further assigned 6 (46.2 %) isolates of F and B2 groups to phylogroup G. *E. coli* belonging to phylogroup G are highly virulent stains and pose a greater threat to neonatal health.

Virulence typing of bacterial pathogens

Selected virulence marker genes were screened using PCR among bacterial pathogens (Table 3). 76.9 % of isolated *S. aureus* showed presence of virulence gene *Luk-PV* which encodes for Pantone-valentine leukocidin toxin, which is a pore forming toxin that creates β -barrel pores in cell membrane of host leukocytes and neutrophils. Genes *clbA*, *clbQ*, *ecpA*, *papC*, *fimH*, *traT* and *aggR* are possessed by *Enterobacteriaceae*. None of the isolated *E. coli* showed presence of genes *clbA* and *clbQ* which encode a cytotoxin named, colibactin, which enables *Enterobacteriaceae* to cause chronic inflammation associated with colorectal cancer [12]. Meanwhile, 30.8 % of isolated *E. coli* and 42.9 % *S. enterica* showed presence of gene *ecpA* which encodes pilli termed as common pillus (ECP), it has potential role in host epithelial cell colonization and may represent a mechanism of adherence in both pathogenic and commensal *Enterobacteriaceae* [13]. 46.2 % of isolated *E. coli* and 42.9 % of *S. enterica* showed presence of gene *papC* which encodes for “PapC protein”, it is present in the outer membrane of *Enterobacteriaceae* where it forms a trans-membrane channel through which pilin subunits are surface localized [14]. 61.5 % of isolated *E. coli* and 42.9 % *S. enterica* showed

Table 2
List of *E. coli* isolates classified into specified phylogroups according to Clermont et al.; 2013, Clermont et al.; 2019.

Number of <i>E. coli</i> isolates	Gene Amplicon					Phylogroup
	<i>chuA</i>	<i>yjaA</i>	<i>TspE4. C2</i>	<i>arpA</i>	<i>YbgD.G</i>	
3 (23.08 %)	-	-	-	-	-	Clade III, IV or V
3 (23.08 %)	+	-	-	-	-	F
3 (23.08 %)	+	-	+	-	-	B2
3 (23.08 %)	-	+	-	-	-	Clade I or II
1 (7.7 %)	+	-	+	-	-	D or E
4 (66.66 %)	-	-	-	-	+	Phylogroup G

Table 3
Molecular analysis of selected virulence/resistance genes based on conventional PCR method of isolated bacterial pathogens.

Virulence gene	<i>Escherichia coli</i>	<i>Salmonella</i> <i>enterica</i>	<i>Citrobacter</i> <i>freundii</i>	<i>Staphylococcus</i> <i>aureus</i>
<i>clbA</i>	0	-	-	-
<i>clbQ</i>	0	-	-	-
<i>ecpA</i>	4 (30.8 %)	3 (42.9 %)	-	-
<i>papC</i>	6 (46.2 %)	3 (42.9 %)	-	-
<i>fimH</i>	8 (61.5 %)	3 (42.9 %)	-	-
<i>traT</i>	7 (53.9 %)	0	-	-
<i>aggR</i>	2 (15.4 %)	3 (42.9 %)	-	-
<i>MecA</i>	-	-	-	29 (74.3 %)
<i>Luk-PV</i>	-	-	-	30 (76.9 %)
<i>MecA</i> and <i>Luk PV</i>	-	-	-	22 (56.4 %)
<i>SltIIB</i>	-	-	9 (47 %)	-
<i>ViaB</i>	-	-	-	-

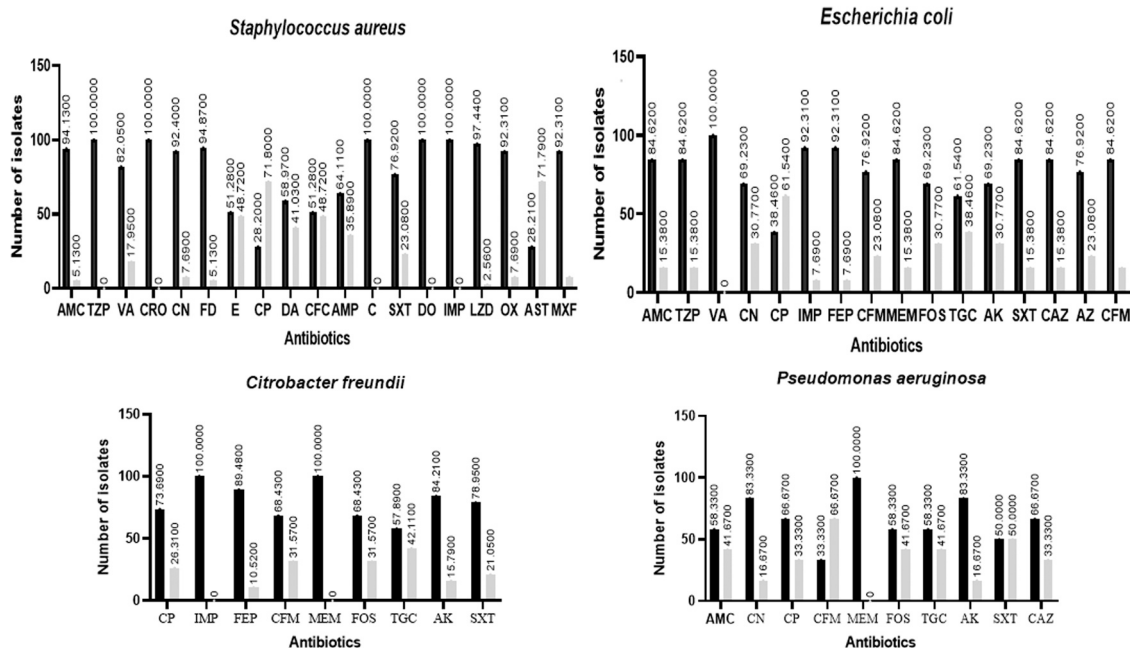


Fig. 1. Resistance and susceptibility of bacterial pathogens, *S. aureus* (n=39), *E. coli* (n=13), *C. freundii* (n=19) and *P. aeruginosa* (n=12), was checked against commonly prescribed antibiotics following bar chart shows number of isolates resistant to antibiotic (Augmentin=AMC, Pepracillin + Tazobactam=TZP, Vancomycin=VA, Ceftriaxone= CRO, Gentamicin=CN, Fusidic acid= FD, Erythromycin=E, Ciprofloxacin=CP, Clindamycin=DA, Cefaclor=CFC, Ampicillin=AMP, Chloramphenicol=C, Co-trimoxazole=SXT, Doxycycline=DO, Imipenem= IMP, Linezolid=LZD, Oxacillin=OX, Penicillin=AST, Moxifloxacin=MXF, Azithromycin=AZ, Cefepime=FEP, Cefixime=CFM, Meropenem=MEM, Fosfomycin=FOS, Tigecycline=TGC, Amikacin=AK and Sulfamethoxazole=SXT).

presence of gene *fimH* which encodes for fimbria-associated pilin domain and a mannose-binding lectin domain, with the binding pocket positioned opposite to the interdomain interface. 53.9 % of isolated *E. coli* showed presence of gene *traT* encoding a lipoprotein which is present in outer membrane and enables *Enterobacteriaceae* to escape complement mediated bacteriolysis [15]. Meanwhile none of *S. enterica* showed presence of gene *traT*. *aggR* is a transcription regulator for different virulence genes of *Enterobacteriaceae*, these genes encode virulent proteins like fimbriae subunit (AAF) and type VI secretion system [16], 15.4 % of isolated *E. coli* and 42.9 % *S. enterica* showed presence of this gene. In addition, Shiga like toxin *SttII*B and *viaB* genes have been seen in certain virulent strains of *C. freundii*. Of the 19 *C. freundii* isolates, 9 (47 %) tested positive for the *SttII*B toxin, however none carried the *viaB* gene.

Antimicrobial susceptibility profiling of bacterial pathogens associated with neonatal sepsis

Phenotypic analysis of antibiotic resistance by Kirby bauer disc diffusion method revealed that bacterial pathogens isolated showed resistance to antibiotics prescribed as first line of treatment including β -lactam drugs. Isolates also depicted resistance against aminoglycosides like gentamycin, except *S. enterica* which was susceptible to all the antibiotics. It was observed that 72 % of the *S. aureus* isolates were resistant to ciprofloxacin (Fig. 1). Therefore, ciprofloxacin is no longer effective in treatment of *S. aureus* infection.

Table 4
Antibiotic resistance and susceptibility profile of bacterial pathogens.

Bacterial species	MDR isolates	Common resistant antibiotic with isolate source		Common susceptible antibiotics
		pus	blood	
<i>Staphylococcus aureus</i>	64.1 %	AST	CP	AMC, TZP, CRO, C, DO, IMP, SXT, LZD & MXF
<i>Citrobacter freundii</i>	31.6 %	FOS, AK & SXT	CFM, TGC, CP	FOS, AK, SXT, MEM & IPM
<i>Pseudomonas aeruginosa</i>	50 %	CP, CFM & SXT	NA	MEM
<i>Escherichia coli</i>	53.8 %	TGC	CP	IMP & FEP
<i>Salmonella enterica</i>	0	NA	NA	MEM, AZ, CFM & CTX

Many bacterial isolates were identified as MDR strains, showing resistance to three or more antibiotics (Table 4). Specifically, 72.2 % of *S. aureus*, 21 % of *C. freundii*, 50 % of *P. aeruginosa* and 24 % *E. coli* were classified as MDR strains. Molecular analysis of methicillin resistance gene *mecA* showed that 74.3 % of isolated *S. aureus* possessed this gene which enabled these strains to produce penicillin binding protein 2a which has less affinity for β -lactam antibiotics.

Meanwhile, 56.4 % *S. aureus* showed presence of both *mecA* and *Luk-PV* genes, depicting merging hospital and community acquired *S. aureus* strains. *Enterobacteriaceae* possessing metallo-beta-lactamases (MBL) are emerging as ceftazidime, cefotaxime, cefepime, imipenem, meropenem, and ampicillin resistant strains [17]. Screening of MBL genes showed that 53.9 % strains of *E. coli* possess GIM gene which encodes for metallo-beta-lactamase and enables bacterial pathogen to develop resistance against β -lactam antibiotics.

Discussion

In Pakistan 1.13–3.8 newborns die each year among 1000 suffering from neonatal sepsis [18]. Profile of bacteria causing neonatal sepsis as well as their antibiotic susceptibility pattern is different in different countries around the world because of lifestyle changes and effective preventive measures taken [17]. This is supported by the observation that bacterial profile of neonatal sepsis is especially varied among developing and developed countries [19]. Considering

the incidence rate of the disease in Pakistan, this study was designed to evaluate the bacterial pathogens frequently associated with neonatal sepsis in a tertiary care setup in Peshawar.

In this study we observed that most prevalent pathogen isolated was *S. aureus* (isolated from 43 % of samples) followed by *C. freundii* (isolated from 21 % of samples), *P. aeruginosa* (isolated from 13 % of samples), *E. coli* (isolated from 15 % of samples) and *S. enterica* (isolated from 8 % of samples). A previous report also shows that etiological organisms of neonatal sepsis were most commonly *Staphylococcus aureus*, coagulase negative *Staphylococci* (CoNS) and *Klebsiella pneumoniae* in under-developed countries [20]. Furthermore, *S. enterica* (8 % infection rate) was isolated only from blood samples while *P. aeruginosa* was only isolated from pus samples of patients who were hospitalized for more than 6 days. These were most likely nosocomial infections with longer hospital stay of patients in contrast to previous reports where *S. enterica* was associated with early onset sepsis (EOS) with transmission from mother to fetus observed (reported in 63 % of the cases) [21]. Meanwhile, patients with *P. aeruginosa* associated sepsis were hospitalized for 0–30 days [22]. Similar to *Salmonella typhi* [23], *E. coli* are also associated with EOS but in this study both *E. coli* and *Salmonella typhi* were associated with LOS.

For virulence typing of *S. aureus* *LukSF-PV* gene was targeted. Pantone-Valentine leucocidin (PVL) is notorious for pus formation in skin [8]. 56.4 % of the isolated *S. aureus* were both *LukSF-PV* and *MecA* positive. Treatment of such strains is difficult [24] as carriage of PVL toxin is thought to be responsible for severe clinical infections caused by community acquired methicillin resistant *S. aureus* (MRSA) [20]. The mechanism of PVL toxicity includes cell lysis which promotes tissue damage [25].

To check pathogenicity of *E. coli* and *S. enterica*, presence of virulence gene was screened. *E. coli* showing presence of these virulence genes is regarded as extra-intestinal pathogenic *E. coli*. We found that *E. coli* isolates were *ecpA* positive (30.8 %), *papC* positive (46.2 %), *fimH* positive (61.5 %) and *traT* positive (53.9 %). Meanwhile *S. enterica* isolates were *ecpA* positive (42.9 %), *papC* positive (42.9 %), *fimH* positive (42.9 %) and *traT* negative. Another study depicted a large number of *E. coli* associated with neonatal sepsis showing presence of *fimH* (90 %), and *traT* (66 %) [26].

E. coli isolates were further selected for phylotyping, as it an efficient way to establish virulence potential and common niche of *E. coli* isolates [27]. The recently proposed quadruplex PCR is highly efficient as it provides a single platform for the detection of eight phylogroups, other cryptic clades and species of *Escherichia* and provides an efficient and cost-effective alternative to Multi-locus Sequence Typing (MLST) [9]. Results of phylogrouping performed for strains of *E. coli* showed that strains belonged to phylogroup F (23 %), B2 (23 %) while 23 % strains belonged to Clade I, II, IV while only 1 (7.7 %) isolate was of phylogroup D or E. While a study conducted in past, concluded that phylogroup B2 was most commonly associated with neonatal sepsis [28]. B2 and F phylogroups are commonly associated with inflammatory bowel disease and are characterized by the expression of *chuA* and *yjaA*, *chuA* virulence genes respectively [29]. It has been observed that multidrug resistant *C. freundii* infection can cause meningitis in newborns although the incidence of brain abscess is not high [30]. We observed a relatively high infection rate of 21 % associated with *C. freundii* in neonatal sepsis cases with number of isolates carrying the Slt-IIIB toxin, an observation that is alarming as Slt-IIIB toxin may confer additional cytotoxic properties to the bacterial cell [31].

Literature has indicated that according to world health organization if a neonate develops signs of neonatal sepsis then gentamicin can be given, another antibiotic that can be given as first line of choice is penicillin derivatives [32]. High bacterial resistance has been reported against first and second line of choice of antibiotics prescribed for the treatment of neonatal sepsis [33]. In past highest

resistance of *E. coli* has been observed against ampicillin (75 %) and ampicillin-sulbactam (52 %) [33]. In another study it was found that *S. aureus* isolated from samples of patients suffering from neonatal sepsis were highly resistant to Augmentin (66.7 %) and clarithromycin (14.3 %) [34]. This study showed highest resistance of *E. coli* strains against sulbactam/ cefoperazone (38.46 %) and tigecycline (38.46 %), while majority isolates of *S. aureus* were resistant to Ciprofloxacin and penicillin (71.80 %). Resistance of *P. aeruginosa* to cefixime (66.7) and levofloxacin (33.3) has also been reported [34], while in our study, maximum number of *P. aeruginosa* isolated from samples of patient showed resistance to Ciprofloxacin (62 %) and cefixime (61.6 %). Highest resistance of *C. freundii* was recorded against tigecycline (42.1 %). While previously, high resistance against amoxicillin (45.5 %) in *C. freundii* has been observed [35]. *S. enterica* was sensitive to the all the antibiotics being tested. In past resistance of gram positive bacteria has been reported against penicillin derivative: ampicillin while resistance against ciprofloxacin was not recorded in highly resistant category [33] but we that most resistant antibiotic observed in different bacterial species was ciprofloxacin.

Conclusions

In our study, *S. aureus* was the most common etiological agent observed for neonatal sepsis while *C. freundii*, *P. aeruginosa*, *E. coli* and *S. enterica* were also isolated from blood and pus samples. Molecular analysis revealed that bacterial pathogens possessed several virulence genes. Majority of isolates were resistant to ciprofloxacin, tigecycline, cefaclor, cefixime, ciprofloxacin and penicillin, while *S. enterica* was sensitive to all the antibiotics tested. We observed that combination of antibiotics showed better results, like Pepracillin + Tazobactam combination (100 % efficacious) in addition to such combination antibiotics carbapenems were also efficacious.

Ethical approval

Study was approved by the COMSATS ethical review board (letter no: CIIT/Bio/ERB/2020/12).

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CRediT authorship contribution statement

ZZ carried out the experimentation and drafted the manuscript, AM and NB carried out the experimentation, ZF, ZN and NR helped in sample collection and contributed to conception and initial clinical data validation, SS and NB provided materials, data validation and manuscript editing, SJ edited the manuscript, designed experiments and approved final manuscript.

Data Availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Conflict of interest

The authors declare no conflict of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2022.12.017.

References

- [1] Mustefa A, Abera A, Aseffa A. Prevalence of neonatal sepsis and associated factors amongst neonates admitted in arbaminch general hospital, arbaminch, southern Ethiopia, 2019. *J Pediatr Neonatal Care* 2020;10(1):1–7.
- [2] Balayan S, Chauhan N, Chandra R, et al. Recent advances in developing biosensing based platforms for neonatal sepsis. *Biosens Bioelectron* 2020;169:112552.
- [3] Sorsa A. Epidemiology of neonatal sepsis and associated factors implicated: observational study at neonatal intensive care unit of Arsi University Teaching and Referral Hospital, South East Ethiopia. *Ethiop J Health Sci* 2019;29:3.
- [4] Ahmed M, Yasrab M, Khushdil A, et al. Neonatal sepsis in a tertiary care hospital: bacteriological profile and its antimicrobial sensitivity. *PAFMJ* 2018;68(6):1654–8.
- [5] Sgro M, Shah PS, Campbell D, et al. Early-onset neonatal sepsis: rate and organism pattern between 2003 and 2008. *J Perinatol* 2011;31(12):794–8. (Dec).
- [6] Humphries R, Bobenchik AM, Hindler JA, et al. Overview of changes to the clinical and laboratory standards institute performance standards for antimicrobial susceptibility testing, M100. *J Clin Microbiol* 2021;59(12). e00213–21.
- [7] Dashti AA, Jadaon MM, Abdulsamad AM, Dashti HM. Heat treatment of bacteria: a simple method of DNA extraction for molecular techniques. *Kuwait Med J* 2009;41(2):117–22.
- [8] Oliveira D, Borges A, Simões M. *Staphylococcus aureus* toxins and their molecular activity in infectious diseases. *Toxins* 2018;10(6):252.
- [9] Clermont O, Christenson JK, Denamur E, et al. The C lermont E scherichia coli phylo-typing method revisited: improvement of specificity and detection of new phylo-groups. *Environ Microbiol Rep* 2013;5(1):58–65.
- [10] Clermont O, Dixit OV, Vangchhia B, et al. Characterization and rapid identification of phylogroup G in *Escherichia coli*, a lineage with high virulence and antibiotic resistance potential. *Environ Microbiol* 2019;21(8). 3107–17.
- [11] Elshamy AA, Aboshanab KM. A review on bacterial resistance to carbapenems: epidemiology, detection and treatment options. *Future Sci OA* 2020;6(3):FSO438.
- [12] Morgan RN, Saleh SE, Farrag HA, et al. Prevalence and pathologic effects of colibactin and cytotoxic necrotizing factor-1 (Cnf 1) in *Escherichia coli*: experimental and bioinformatics analyses. *Gut Pathog* 2019;11(1):1–8.
- [13] Munhoz DD, Nara JM, Freitas NC, et al. Distribution of major pilin subunit genes among atypical enteropathogenic *Escherichia coli* and influence of growth media on expression of the *ecp* operon. *Front Microbiol* 2018;9:942.
- [14] Yen MR, Peabody CR, Partovi SM, et al. Protein-translocating outer membrane porins of Gram-negative bacteria. *Biochim Et Biophys Acta (BBA)-Biomembr* 2002;1562(1–2):6–31.
- [15] Zückert WR. Secretion of bacterial lipoproteins: through the cytoplasmic membrane, the periplasm and beyond. *Biochimica et Biophysica Acta (BBA)-Mol Cell Res* 2014;1843(8):1509–16.
- [16] Dias RC, Tanabe RH, Vieira MA, et al. Analysis of the virulence profile and phenotypic features of typical and atypical enteroaggregative *Escherichia coli* (EAEC) isolated from diarrheal patients in Brazil. *Front Cell Infect Microbiol* 2020;10:144.
- [17] Gautam S, Bhattarai NR, Rai K, et al. Detection of bla NDM-1 encoding impenemase among the imipenem-resistant gram-negative bacilli isolated from various clinical samples at a tertiary care hospital of eastern Nepal: a descriptive cross-sectional study. *Int J Microbiol* 2020;2020.
- [18] Ahmed M, Yasrab M, Khushdil A, Qamar K, Ahmed Z. Neonatal sepsis in a tertiary care hospital: bacteriological profile and its antimicrobial sensitivity. *Pak Armed Forces Med J* 2018;1(6):1654–8.
- [19] Stoll BJ, Puopolo KM, Hansen NI, et al. Early-onset neonatal sepsis 2015 to 2017, the rise of *Escherichia coli*, and the need for novel prevention strategies. *JAMA pediatrics*. 2020 Jul 1;174(7):e200593–.
- [20] Rafi MA, Miah MM, Wadood MA, et al. Risk factors and etiology of neonatal sepsis after hospital delivery: a case-control study in a tertiary care hospital of Rajshahi, Bangladesh. *PloS One* 2020;15(11):e0242275.
- [21] Shrestha N, Shrestha D. *Salmonella* sepsis presenting as early onset Neonatal sepsis. *J Nepal Paediatr Soc* 2018;38(1).
- [22] Harnaen E, Doctor TN, Malhotra A. Characteristics of *pseudomonas aeruginosa* infection in a tertiary neonatal unit. *Int J Pediatr Res* 2015;1:010.
- [23] Walser S, El Moussawi F, Sire JM, et al. Late-onset sepsis due to *Salmonella typhi* in a preterm infant in a French neonatal unit. *Arch De Pédiatrie* 2021;28(1):93–5.
- [24] Dong Y, Glaser K, Speer CP. New threats from an old foe: methicillin-resistant *Staphylococcus aureus* infections in neonates. *Neonatology* 2018;114(2). 127–34.
- [25] Sina H, Ahoyo TA, Moussaoui W, et al. Variability of antibiotic susceptibility and toxin production of *Staphylococcus aureus* strains isolated from skin, soft tissue, and bone related infections. *BMC Microbiol* 2013;13(1):1–9.
- [26] Munkhdelger Y, Gunregjav N, Dorjpurev A, et al. Detection of virulence genes, phylogenetic group and antibiotic resistance of uropathogenic *Escherichia coli* in Mongolia. *J Infect Dev Ctries* 2017;11(01):51–7.
- [27] Gordon DM, Clermont O, Tolley H, et al. Assigning *Escherichia coli* strains to phylogenetic groups: multi-locus sequence typing versus the PCR triplex method. *Environ Microbiol* 2008;10(10):2484–96.
- [28] Cole BK, Ilik M, McCloskey CB, Chavez-Bueno S. Antibiotic resistance and molecular characterization of bacteremia *Escherichia coli* isolates from newborns in the United States. *PLoS One* 2019;14(7):1–17.
- [29] Fang X, Monk JM, Mih N, et al. *Escherichia coli* B2 strains prevalent in inflammatory bowel disease patients have distinct metabolic capabilities that enable colonization of intestinal mucosa. *BMC Syst Biol* 2018;12(1). 1–0.
- [30] Chen D, Ji Y. New insights into *Citrobacter freundii* sepsis in neonates. *Pediatr Int* 2019;61(4):375–80.
- [31] Schmidt H, et al. Shiga-like toxin II-related cytotoxins in *Citrobacter freundii* strains from humans and beef samples. *Infect Immun* 1993;61(2):534–43.
- [32] Korang SK, Safi S, Gluud C, et al. Antibiotic regimens for neonatal sepsis-a protocol for a systematic review with meta-analysis. *Syst Rev* 2019;8(1):1–3.
- [33] Tessema B, Lippmann N, Knüpfer M, et al. Antibiotic resistance patterns of bacterial isolates from neonatal sepsis patients at university hospital of Leipzig, Germany. *Antibiotics* 2021;10(3):323.
- [34] Atif M, Zia R, Malik I, et al. Treatment outcomes, antibiotic use and its resistance pattern among neonatal sepsis patients attending Bahawal Victoria Hospital, Pakistan. *PLoS One* 2021;16(1):e0244866.
- [35] Khadka SB, Thapa B, Mahat K. Nosocomial *Citrobacter* infection in neonatal intensive care unit in a hospital of Nepal. *J Nepal Paediatr Soc* 2011;31(2):105–9.